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Optimal Booster Chlorination Scheduling in WDS under Uncertainty: A Robust Counterpart Approach

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ABSTRACT

To ensure high-quality water in water distribution systems (WDS), disinfectant, generally chlorine, is injected into the system. However, this chlorine limits should be kept within acceptable levels. The higher limit is enforced to control the disinfection by-products, whereas lower limit is enforced to ensure that the water is organic contaminant free. The chlorine reaction rate within the pipe systems varies depending on the type of pipe, age, water PH level, temperature, and many other factors. This hinders the accurate water quality modelling in water distribution systems which in turn affects the optimal amount of booster injection rate. To minimize the risk of the uncertainty in chlorine reaction rate, current work suggests a robust counterpart reformulation of the optimal booster chlorination scheduling problem considering the rate of reaction of chlorination as uncertain. The proposed reformulation is tested on a benchmark WDS, and the results are compared with deterministic case. The results indicated that with an increase in the protection level the injection mass increased and with very large protection levels, the formulation resulted in no solution. The proposed reformulation of the traditional approach can be applied to any WDS and obtain optimal scheduling with appropriate protection levels.

INTRODUCTION

Drinking water distribution systems are an essential part of the urban development. The water distribution systems aim to provide high quality water at required pressures and also satisfy the consumer demands. With the rapid increase in the urban standards and urban population these systems are continuously stressed. As huge populations are dependent on these systems optimal design and operation of these systems are essential. To ensure adequate water quality to customers chlorine is used as the disinfectant. During the water transfer from source to customers through pressured pipes, the injected chlorine reacts with inorganic and organic contaminants through complex physical, chemical and also biological mechanisms. The chlorine concentration decays during to water transfer in WDS, this leads to reduced chorine concentrations at the consumer nodes, which can affect the future control of growth in micro-organisms (Kang et al. 2009a). This made to employ minimum chlorine concentration requirement at the end of the customer node. On the other hand, high levels of chlorine can cause several disinfection by-products that can cause serious health problems (Wang et al. 2018). In

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addition to this, high chlorine concentration leads to strong taste and odour which is unpleasant to consume. These problems encouraged the WDS designers to ensure maximum chlorine levels in addition to minimum chlorine levels. The safe range of chlorine concentration should consider the trade-off between controlling the pathogen growth and reduction of disinfection by-products (DBPs) to prevent long-term health hazards(Boccelli et al. 1998).The safe range of chlorine concentration varies from country to country guidelines general limits of chlorine concentration are considered from 0.2 mg/l up to 4.0 mg/l (Goyal and Patel 2015, 2017; Köker and Altan-Sakarya 2015). In order to maintain the chlorine concentration within the limits, booster chlorination stations are strategically placed and are optimally operated.

Optimization of WDS initially focused on the design of the system focussing on meeting the hydraulic and water quality requirements. Later the research focuses shifted to optimizing the operational strategies and later a combination of both design and operational optimization. The booster chlorination dosage optimization falls under operational optimization problem. Numerous researchers have explored multiple optimization strategies to obtain optimal dosage scheduling of booster chlorination systems. Linear programming optimization strategy was used with the objective of minimizing the chlorine injected mass while ensuring the residual chlorine is within the safe bounds(Boccelli et al. 1998; Tryby et al. 1999). A linear least square model was used to minimize the deviations from the limits by (Propato and Uber 2004). Multi-objective optimization models were also used to minimize the chlorine mass as well as maximize the volume of water supplied that is within the limits of residual chlorine using genetic algorithm (Prasad et al. 2004). Another extension of this multi-objective problem was to include the minimization of cost of pumping by (Ostfeld and Salomons 2006). Many other variations of the optimal booster chlorination problem is solved using single and multi-genetic algorithms, swarm intelligence algorithms(Chu et al. 2008; Harmant et al. 2000; Ohar and Ostfeld 2014; Prasad et al. 2004; Wang et al. 2010).

Uncertainty is a constraint which is hindering the reliability of the deterministic optimal strategies of both design and operations of WDS. To overcome this difficulty and to obtain more reliable optimal strategies researchers reformulated the traditional deterministic problem formulations to include uncertainty. Initially the research focused on solving the design and operational problems considering hydraulic uncertainty using chance constraints, approximation techniques, and other stochastic optimization methods [(Chung et al. 2009; Hwang et al. 2018; Jung et al. 2012; Kang et al. 2009b; Lansey et al. 1989; Seifollahi-Aghmiuni et al. 2013)]. Due to high computational time requirement for stochastic techniques, research slowly moved towards using non-probabilistic optimization under uncertainty techniques [(Boindala et al. 2022; Boindala and Ostfeld 2022; Pankaj et al. 2022; Perelman et al. 2013; Schwartz et al. 2016)].

Most of the previous research mainly focused on hydraulic uncertainty like uncertainty in demands, pipe roughness, pipe diameter, etc. Very little research considered uncertainty in water quality parameters. (Pasha and Lansey 2005, 2010) examined the effect of few water quality uncertain parameters like bulk and wall reaction coefficients, and pipe diameters in the distribution system's water quality analysis. Optimal booster chlorination problem is also effected by uncertainty in bulk decay rate of chlorine. (Köker and Altan-Sakarya 2015)used chance constraint programming to formulate and solve the optimal booster chlorination scheduling problem. Wang and Zhu proposed inexact left-hand side two-stage chance-constrained programming for booster optimization in water distribution system. There have not been any studies that considers non-probabilistic optimization techniques like robust optimization to solve the optimal booster chlorination problem under uncertainty.

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ROBUST OPTIMIZATION

Robust optimization (RO) is a technique independently developed by (Ben-Tal and Nemirovski 1998, 1999; el Ghaoui and Lebret 1997). Recently a textbook was published by Ben-Tal that compiles the research and applications of RO in various sectors in science and engineering (Ben-Tal et al. 2009). This approach provides a more conservative solutions as compared earlier worst-case methods by bounding the uncertain variables to a standardised uncertainty sets. The key advantage of these techniques is their tractability within a specified uncertainty sets like box or ellipsoid. These techniques have wide applications like portfolio management, optimal shortest path, optimal facilities locations, optimal traffic control, etc. These techniques have also been used in solving WDS problems (Boindala et al. 2022; Boindala and Ostfeld 2022; Perelman et al. 2013; Schwartz et al. 2016).

RO- A short tutorial:

Consider the following linear programming problem:

$$\begin{aligned} &\min_{x \in X} C^T x \\ &s.t\ Ax \geq b \end{aligned} \tag{1}$$

Where $x \in X \subseteq \mathbb{R}^n$ is a vector of decision variables, $c \in \mathbb{R}^n$ is the cost coefficient vector associated with the objective function, $A \in \mathbb{R}^{mXn}$ is the constraint coefficient matrix, and $b \in \mathbb{R}^m$ is a right-hand side inequality constraint vector.

In a typical LP problem, the vectors c, b and the matrix A are deterministic and we solve the problem and obtain optimal solution. In the RO problem, we consider some/all of these parameters as uncertain, yet lying within a specified set. This set is called the uncertainty set which defines the limits on uncertainty in which the robust solution to this problem is immune to. In this RO approach the uncertain parameters of the LP problem is converted into a constraint that reflects the immunity within the uncertainty set. The converted tractable constraint of the uncertain constraint is called the robust counterpart(Baron et al. 2011).

Box Uncertainty:

Let us assume that the constraint coefficient matrix A as uncertain, i.e. every element in the matrix A $(\tilde{a}_{ij}, i = 1,2,3 \ldots, n, j = 1,2,3, \ldots, m)$ is uncertain where the coefficient is bounded within a Box uncertainty set (U_B). Let us define the uncertainty set as: $U_B \equiv \{\tilde{a}: |\tilde{a} - \bar{a}| \leq \epsilon |\hat{a}|\}$ here $\bar{a}$ is the mean and $\hat{a}$ is the deviation from the mean. The uncertainty can be removed by replacing the uncertain constraint with a minimization counterpart constraint which can be written as follows.

$$\min_{x \in X} C^T x \tag{2}$$

$$\min_{\tilde{a} \in U_B} \left\{ \sum_{i=1}^{n} \tilde{a}_{ij} x_i \right\} \geq b_j \Leftrightarrow \min_{\tilde{a}: |\tilde{a} - \bar{a}| \leq \epsilon |\hat{a}|} \left\{ \sum_{i=1}^{n} \tilde{a}_{ij} x_i \right\} \geq b_j \tag{3}$$

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Given the structure of U_B, the optimal solution of the minimization of the LHS of the constraint can be written as

$$\min_{\tilde{a}:|\tilde{a}-\bar{a}|\leq\epsilon|\hat{a}|}\left\{\sum_{i=1}^{n}\tilde{a}_{ij}x_i\right\}\geq b_j => \sum_{i=1}^{n}\bar{a}_{ij}x_i-\epsilon\sum_{i=1}^{n}\left|\hat{a}_{ij}\right|\left|x_{ij}\right|\geq b_j \tag{4}$$

This reformulated constraint is robust within the uncertainty set and with this new constraint the uncertain LP can be easily be solved(Baron et al. 2011).

OPTIMAL BOOSTER DISINFECTION PROBLEM

The quality of water from WDS is generally quantified by its residual chlorine concentration. During the design phase, the water quality is estimated through water quality models. The water quality models like (EPANET) needs parameters like bulk reaction coefficient of chlorine as input, which is generally governed by the contaminants present in the water, age and material of the pipe, pH and temperature of the water and many more external parameters which are quite difficult to predict accurately. This makes the reaction coefficient parameter as uncertain. This uncertainty in the reaction coefficient makes it difficult to plan the optimal rate of disinfectant mass injection to maintain the water quality requirements over a long operational period.

The optimization problem can be defined as follows:

We modify the linear programming model proposed by (Boccelli et al. 1998). The disinfectant is injected at n nodes, and there are N demand nodes. The decision variable is an n-dimensional column vector which represents the amount of disinfectant mass injected into the system from all the n nodes. It is denoted by $\bar{m}\in\mathbb{R}^n$. The constraint enforces the amount of disinfectant mass at each demand node to be within the given upper and lower bounds. $C^U$and C^L are the upper and lower limits of residual disinfectant concentrations. B is the unit response matrix obtained from unit disinfectant mass injected into the system. It is a matrix of dimension $N\times n$. The detailed explanation of obtaining the response matrix can be found in Bocelli et al. The objective function that we try to minimize is $f(m)$, the total disinfectant mass injected into system.

Objective:

$$min\, f(m)=\sum_{i=1}^{n}m_i \tag{5}$$

Constraints:

$$C^L\leq\sum_{i=1}^{n}B_{ij}m_i\leq C^U, j=1,2,\dots,N \tag{6}$$

B_{ij} is the element in the row j and column i of the unit response matrix B. When the reaction coefficient is known, the response matrix is certain, and the problem becomes a standard LP problem.

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The response matrix is unique for every decay rate coefficient of chlorine, when the reaction coefficient is unknown, the response matrix is also unknown. When the decay rate coefficient varies within a box uncertainty set $\widetilde{K_B} \in [K_B^L, K_B^U]$, the uncertain response matrix also varies within a box uncertainty set $\tilde{B} \in [B_L, B_U]$. With this uncertain response matrix, the above formulation should be modified so that the new formulation will find the disinfection mass vector (mass vector) that is feasible for all possible values of the uncertain response matrix set because of the uncertain reaction coefficient. That is,

Objective:

$$min\, f(m) = \sum_{i=1}^{n} m_i \tag{7}$$

Constraints:

$$C^L \leq \sum_{i=1}^{n} \tilde{B}_{ij} m_i \leq C^U, \forall \tilde{B}_{ij} \in U, j = 1,2, \dots, N \tag{8}$$

$\tilde{B}_{ij}$ is the uncertain unit response matrix whose value can be any value in the set U. A feasible mass vector will satisfy the constraint for all possible values of $\tilde{B}_{ij} \in U$ for each demand node $j = 1,2, \dots, N$. This problem is intractable because of the infinite number of constraints and solving this requires a reformulation. This is done using the techniques of RO and the resulting problem is called the Robust Counterpart (RC).

The reformulation is done as follows. Consider one constraint for a particular demand node j. We will explain this reformulation for this constraint, and this can be easily extended to any constraint. Consider the infinite number of chlorine upper bound constraint. We rewrite the constraint in the vector form, and $\tilde{B}_j$ now denotes the row j of the unit response matrix. It is a row vector while the mass vector ($\overline{m}$) is a column vector. Both are of length n. Since we consider a box uncertainty set, U_j can be written as $U_j = [\overline{B_{L_j}}, \overline{B_{U_j}}]$.

$$\tilde{B}_j \overline{m} \leq C^U, \forall B_j \in U_j \tag{9}$$

If this constraint is satisfied by some $\overline{m}$ then the largest value of the left-hand side should be at most C^U, else there exists at least one value of $\tilde{B}_j$ such that the left-hand side is strictly greater than C^U and the problem is infeasible. Thus, the above constraint is satisfied if

$$\max_{B_j \in U_j} \tilde{B}_j \overline{m} \leq C^U \tag{10}$$

Now we optimize the left-hand side. For this first we write the Lagrangian to convert the constrained optimization to an unconstrained optimization. Thus, we get

$$\begin{aligned} &\max_{\tilde{B}_j} \tilde{B}_j \overline{m} - \left(\tilde{B}_j - \bar{B}_{U_j}\right) \alpha_j - \left(\overline{B_{L_j}} - \tilde{B}_j\right) \beta_j\,, \\ &=> \max_{\tilde{B}_j} \tilde{B}_j (\overline{m} - \alpha_j + \beta_j) + (\bar{B}_{U_j} \alpha_j - \overline{B_{L_j}} \beta_j) \end{aligned} \tag{11}$$

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where α_j and β_j are dual variables and are column vectors of length n. The solution is ∞ if $\overline{m} - \alpha_j + \beta_j \neq 0$. That is the problem is infeasible if $\overline{m} - \alpha_j + \beta_j \neq 0$. For the problem to be feasible $\overline{m} - \alpha_j + \beta_j$ should be equal to 0, and then the maximum is $\bar{B}_{U_j}\alpha_j - \overline{B_{L_j}}\beta_j$. Thus, the original infinite chlorine upper bound constraint is satisfied if the maximum $\bar{B}_{U_j}\alpha_j - \overline{B_{L_j}}\beta_j$ is at most C^U. Thus, the original constraint is replaced with three constraints:

$$\overline{m} - \alpha_j + \beta_j = 0 \tag{12}$$

$$\bar{B}_{U_j}\alpha_j - \overline{B_{L_j}}\beta_j \leq C^U \tag{13}$$

$$\alpha_j, \beta_j \geq 0 \tag{14}$$

Similarly, we treat the infinite number of chlorine lower bound constraints, and we get the following three constraints for some demand node j.

$$-\bar{B}_{U_j}\gamma_j + \overline{B_{L_j}}\delta_j \geq C^L \tag{15}$$

$$-\overline{m} - \gamma_j + \delta_j = 0 \tag{16}$$

$$\gamma_j, \delta_j \geq 0 \tag{17}$$

γ_j and δ_j are dual variables and are column vectors of length n. Combining everything we get the tractable robust counterpart below.

Objective:

$$min\ f(m) = \sum_{i=1}^{n} m_i \tag{18}$$

Constraints:

$$\overline{m} - \alpha_j + \beta_j = 0, j = 1,2, \dots, N \tag{19}$$

$$\overline{U}_j\alpha_j - \overline{L}_j\beta_j \leq C^U, j = 1,2, \dots, N \tag{20}$$

$$-\overline{m} - \gamma_j + \delta_j = 0, j = 1,2, \dots, N \tag{21}$$

$$-\overline{U}_j\gamma_j + \overline{L}_j\delta_j \geq C^L, j = 1,2, \dots, N \tag{22}$$

$$\alpha_j, \beta_j, \gamma_j, \delta_j \geq 0, j = 1,2, \dots, N \tag{23}$$

$$\overline{m} \geq 0 \tag{24}$$

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CASE STUDY

The above formulation, explained as a flowchart in figure (1) is applied on the Net-1 example of EPANET. The lower and upper bounds of chlorine concentration are considered to be 0.2mg/l and 1mg/l("Drinking Water Systems | Ministry of Health"). Four different combination of three booster locations are considered for the study as shown in the figure. Four different sensor locations are considered for measuring the residual chlorine concentration for 24 hours measured at every 1-hour intervals. The booster chlorination injection and the measurements are taken after 480hr hydraulic extended period simulation. For each combination, the total mass injected is calculated for four different uncertainty sets of decay rate coefficients {[-0.5], [-0.4, -0.6], [-0.3, -0.7], [-0.2, -0.8]}. The objective of the problem is to obtain the optimal total amount of chlorine injection into system to ensure least deviation from the limits. The results shown in the figure (2-5) shows the optimal total amount of chlorine mass injected (mg/min) [in bar graph] and also the corresponding maximum deviation from the chlorine bounds (mg/l) [Line graph] for all the four cases with different booster chlorination stations.

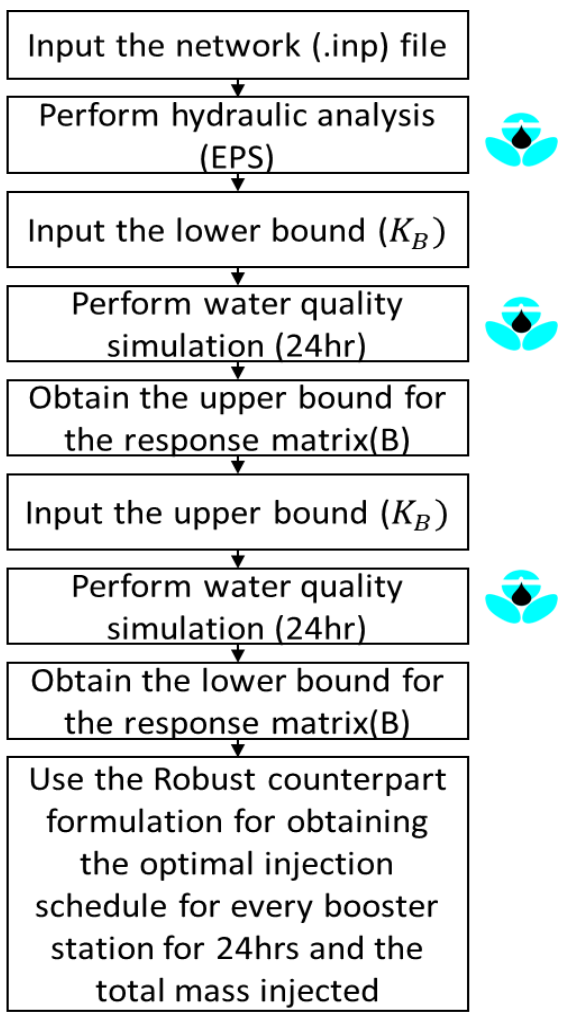

Figure 1. Flowchart of the methodology

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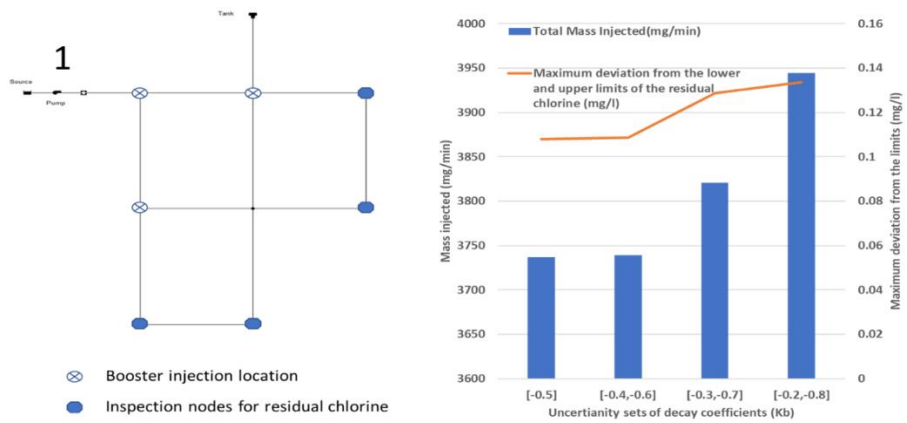

Figure 2. Figure showing the results obtained for case-1 with booster locations at nodes (11,12,21) for Net-1 example of EPANET

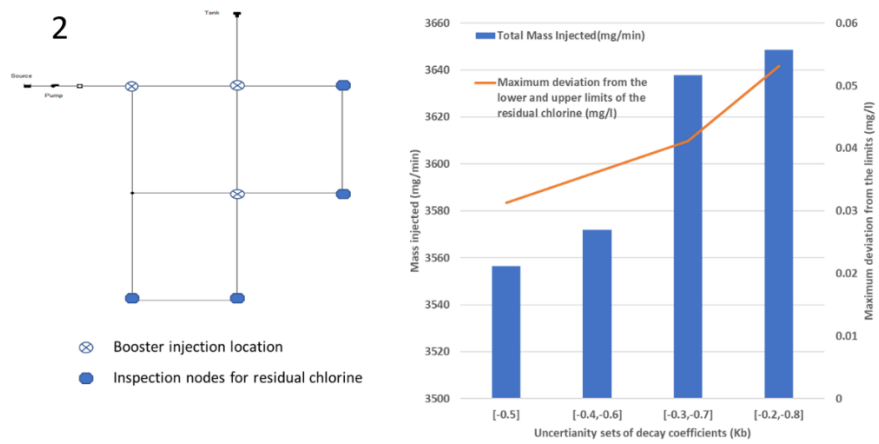

Figure 3. Figure showing the results obtained for case-2 with booster locations at nodes (11,12,22) for Net-1 example of EPANET

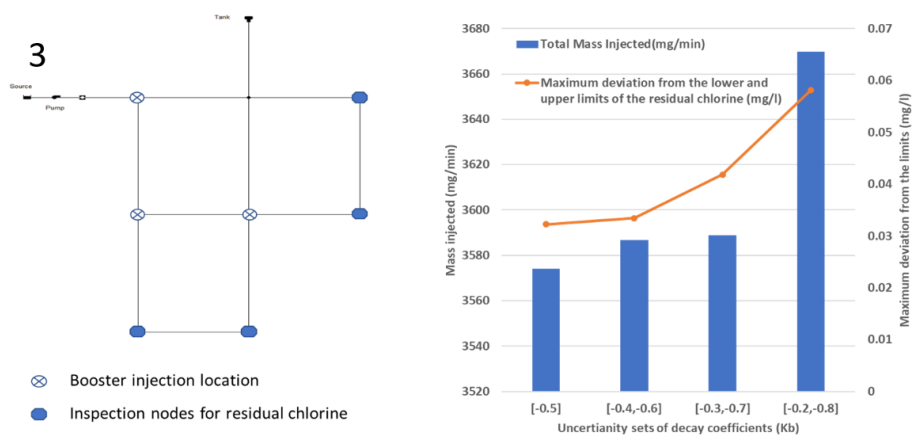

Figure 4. Figure showing the results obtained for case-3 with booster locations at nodes (11,21,22) for Net-1 example of EPANET

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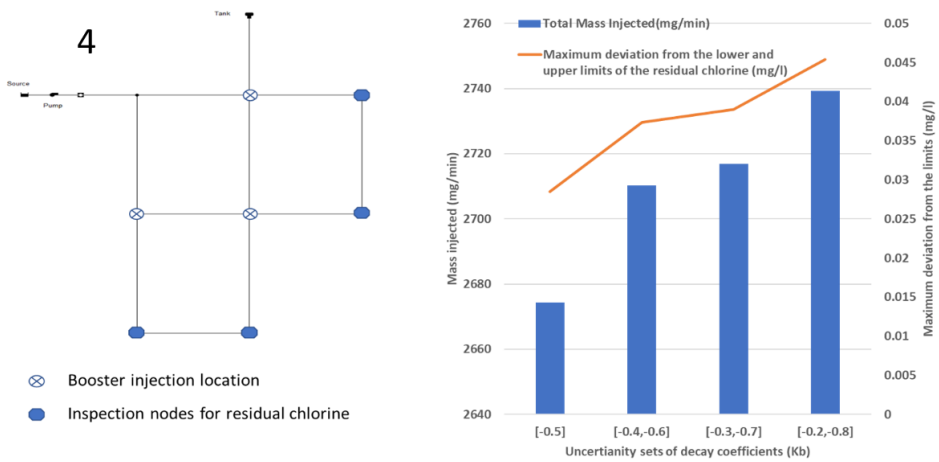

Figure 5. Figure showing the results obtained for case-4 with booster locations at nodes (12,21,22) for Net-1 example of EPANET

CONCLUSIONS

Current study explores robust counterpart approach to optimize the booster chlorination optimization problem considering uncertainty in bulk chlorine reaction rate. Four different uncertainty sets of the reaction rate are compared. The results show with increase in the uncertainty set size, the total mass injected is increased. Four different combinations of booster chlorination stations are considered. This is an introductory study to explore the uses of robust counterpart in booster chlorination optimization problem. The major advantage of this method over standard chance constraint formulation is the computational time, the advantage will be significant when larger networks are considered which can be the future scope of the study. Box uncertainty sets produces conservative solutions, other types of uncertainty sets like ellipsoidal uncertainty sets can be explored as a future scope. The problem can be coupled with other design or operational objective for a multi-objective study.

ACKNOWLEDGMENTS

This research was supported by a grant from the United States–Israel Binational Science Foundation (BSF).

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